Tutorial – 13

Department of Physics **Shiv Nadar U**

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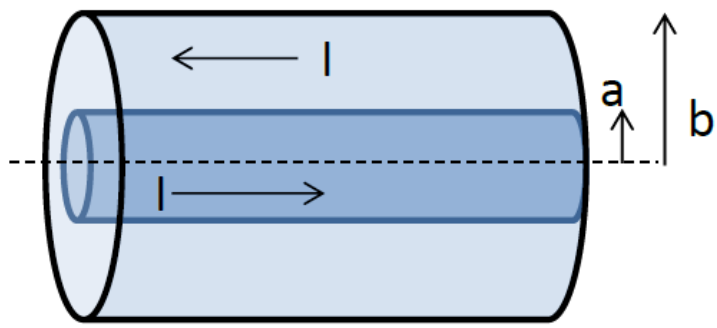
1. A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating magnetic susceptibility χ_m. A current I flows down the inner conductor snd returns along the in each case, the current distributes itself uniformly over the surface. Find the magnetic region between the tubes. Further, calculate the magnetization and the bound currents, that (together, of course, with the free currents) they generate the correct field.



2. A current I flows down a long straight wire of radius 'a'. If the wire is made of linear mate of Aluminium) with susceptibility χ_m, and the current is distributed uniformly, what is the m a distance 's' from the axis? Find all the bound currents. What is the net bound current fl the wire?

3. An infinitely long cylinder, of radius R, carries a frozen-in magnetization, parallel to the axi

$$\boldsymbol{M} = ks\hat{\boldsymbol{z}}$$

where 'k' is a constant and 's' is the distance from the axis; there is no free current anywhe magnetic field inside and outside the cylinder by two different methods:

(a) Locate all the bound currents and calculate the fields they produce.

(b) Use Ampere's law in H $\oint \boldsymbol{H} \cdot \boldsymbol{dl} = \mu_0 I_{f,enc}$ to find **H**, and then get **B** from the equation

$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$$

Notice that this method is much faster, and avoids any explicit reference to bound currents.

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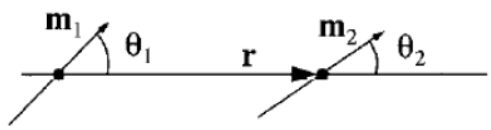
4. (a) Shown that the interaction energy of two magnetic dipoles separated by distance r is given by

$$U = \frac{\mu_0}{4\pi}\frac{1}{r^3}[\boldsymbol{m_1}.\boldsymbol{m_2} - 3(\boldsymbol{m_1}.\hat{\boldsymbol{r}})(\boldsymbol{m_2}.\hat{\boldsymbol{r}})]$$



(b) Express your answer in terms of angles $\theta_1$and θ_2 as shown in figure and use the result to find the stable configuration that the two dipoles would adopt if held a fixed distance apart, but left free to rotate. Explain the origin of ferromagnetism classically from the result.